LIMITING SPEED GOVERNOR

No. 7—IDLE SPEED AND BUFFER SCREW SETTING

Adjustment No. 7

With engine running at idle speed, observe engine RPM at tachometer, then loosen locknut (42) and thread idle adjusting screw (43) in to increase or out to decrease until 385 engine RPM is obtained (fig. 48). Tighten locknut.

With engine running at idle speed, thread buffer screw (46) in until "surge" or "roll" of engine just disappears, but not higher than 400 RPM. Variation from this speed will affect fuel modulating action. Do not raise engine speed more than 20 RPM with the buffer screw, otherwise it may not be possible to stop the engine. Hold screw (46) and tighten locknut (fig. 49). Stop and start engine several times to be sure of its stopping.

INSTALL REMOVED PARTS

1. Install and tighten governor adjustment cover (44).
2. Connect throttle control rod to throttle control lever at governor cover.
3. Connect linkage to engine stop lever at governor cover.

LIMITING SPEED GOVERNOR

GENERAL

Requirements on the engine continually vary, due to fluctuating loads, therefore some means must be provided to control the amount of fuel required for idling, intermediate, and full throttle operation. A no fuel position in a cam for stopping the engine is used on trucks. To accomplish this control, a governor is inserted in the linkage between the manual throttle control and the fuel injectors.

In the governor, force is transmitted from hinged, mechanical rotating weights through the governor operating mechanism to the injector control racks, which are connected to the governor by a link. Forces created by the rotating weights are counterbalanced by springs in the governor, tension of which determines engine speed.

This governor has the following characteristics:

(a) Controls the engine idling speed.

(b) Limits maximum engine speed.

(c) Manual control between maximum engine speed and idling.

(d) Permits return of injectors to no fuel position at speeds above idling range.

NOTE: All illustrations in this section show the governor for LA engine models, unless otherwise specified.

DESCRIPTION

The limiting speed governor (figs. 51, 52 and 55) incorporates two high speed and two low speed weights; each set of weights having a fixed stop for the inner and outer positions. A high and low speed spring is also employed in this governor; the former works in conjunction with the high speed weights to limit the maximum engine speed, the latter works in conjunction with both the low and high speed weights to control the engine idling speed.

The motion produced by the travel of the governor weights, between their inner and outer positions, is transmitted to the injector racks by a system of cams and levers on which can be superimposed a manual control.

The governor, which is mounted at front of blower as shown in fig. 50, is divided into two main assemblies embodied in separate housings. These assemblies are: Governor control housing and governor weight and lower housing, and the cover.

The two sets of weights are carried on a horizontal shaft inside governor weight housing. Weight carrier shaft is mounted on an annular ball bearing at one end and opposite end is supported inside and driven by hollow blower upper rotor shaft. The blower end of governor shaft is serrated and engages with corresponding serrations inside blower shaft, which drives the governor shaft and weights.

Control mechanism transmits motion of governor weights to the injector racks. This mechanism consists of a vertical shaft mounted inside a housing, with a fork or yoke fixed at lower end, an operating lever fixed at upper end, and a high and low speed governing spring with suitable adjustments. The vertical shaft is mounted on an annular ball bearing at upper end and a bushing at lower end.

The motion of the governor weights is transmitted to the vertical shaft through a movable riser on the weight carrier shaft and the fork on the lower end of the vertical shaft. This motion is, in turn, transmitted to the injector control tube by means of the